INCH-POUND
MIL-DTL-24643/36G
27 November 2023
SUPERSEDING
MIL-DTL-24643/36F(SH)
w/INT. AMENDMENT 1
11 July 2011
MIL-DTL-24643/36F
1 October 2009

DETAIL SPECIFICATION SHEET

# CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LS3SWU (INCLUDING VARIATION LS3SWUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS3SWU and LS3SWUS (double overall shield).

## Construction (Watertight)

| | | |
|---|---|---|
| First | - | Copper conductor, coated or uncoated, 18 AWG size, class B stranding (7 by 0.0152 inches). |
| Second | - | Separator may be used at manufacturer's option where required to provide free-stripping insulation. |
| Third | - | Thermoset insulation. Nominal wall thickness of 0.023 inch; wall thickness may vary from nominal as necessary, in order for the completed cable to meet the specified electrical requirements. Colored insulation, one white, one black, and one red conductor for each triad. |
| Fourth | - | Three conductors (one black, one white, and one red) cabled together with a nominal lay of 3.5 inches to form a triad. |
| Fifth | - | An optional binder. |
| Sixth | - | Braided shield of 34 or 36 AWG, coated or uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent. |
| Seventh | - | Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape, plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2. |
| Eighth | - | The specified number of triads (see table I or table II) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly. |



Distribution Statement A. Approved for public release: distribution is unlimited.

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Ninth - An optional binder.

Construction for Variant Type LS3SWU

Tenth - Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Construction for Variant Type LS3SWUS

Tenth - Braided shield of tin-coated copper.

Eleventh - Spirally wrapped, 0.002-inch-thick polyethylene terephthalate (polyester film) separator tape with 25 percent minimum overlap.

Twelfth - Braided shield of tin-coated copper.

Thirteenth - An optional binder.

Fourteenth - Cross-linked polyolefin jacket. See table II for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details (unshielded).

| Military part no. M24643/36 | Type and size | Number of triads | Cable jacket thickness min. avg. (inch) | Overall diameter (inches) | | Hydrostatic leakage (cubic inches) |
|---|---|---|---|---|---|---|
| | | | | min. | max. | |
| -01UN | LS3SWU-3 | 3 | 0.040 | 0.610 | 0.655 | 1.00 |
| -02UN | LS3SWU-7 | 7 | 0.040 | 0.880 | 0.940 | 1.50 |
| -03UN | LS3SWU-10 | 10 | 0.050 | 1.100 | 1.180 | 2.50 |
| -04UN | LS3SWU-14 | 14 | 0.050 | 1.200 | 1.280 | 3.00 |
| -05UN | LS3SWU-19 | 19 | 0.050 | 1.370 | 1.450 | 4.00 |
| -06UN | LS3SWU-24 | 24 | 0.050 | 1.640 | 1.760 | 5.00 |
| -07UN | LS3SWU-30 | 30 | 0.050 | 1.760 | 1.860 | 6.00 |
| -08UN | LS3SWU-37 | 37 | 0.060 | 1.890 | 1.940 | 8.00 |
| -09UN | LS3SWU-44 | 44 | 0.060 | 2.140 | 2.240 | 10.00 |



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TABLE II. Details (shielded).

| Military part no. M24643/36 | Type and size | Number of triads | Cable jacket thickness min. avg. (inch) | Overall diameter (inches) | | Hydrostatic leakage (cubic inches) |
|---|---|---|---|---|---|---|
| | | | | min. | max. | |
| -01UD | LS3SWUS-3 | 3 | 0.040 | 0.675 | 0.710 | 1.00 |
| -02UD | LS3SWUS-7 | 7 | 0.040 | 0.940 | 1.000 | 1.50 |
| -03UD | LS3SWUS-10 | 10 | 0.050 | 1.160 | 1.240 | 2.50 |
| -04UD | LS3SWUS-14 | 14 | 0.050 | 1.260 | 1.340 | 3.00 |
| -05UD | LS3SWUS-19 | 19 | 0.050 | 1.430 | 1.510 | 4.00 |
| -06UD | LS3SWUS-24 | 24 | 0.050 | 1.700 | 1.820 | 5.00 |
| -07UD | LS3SWUS-30 | 30 | 0.050 | 1.820 | 1.920 | 6.00 |
| -08UD | LS3SWUS-37 | 37 | 0.060 | 1.950 | 2.050 | 8.00 |
| -09UD | LS3SWUS-44 | 44 | 0.060 | 2.200 | 2.300 | 10.00 |

REQUIREMENTS:

Qualification required.

INSPECTION:

**Basic electricals:**

| | |
|---|---|
| Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.) | 7.06 |
| Voltage withstand (volts, root mean square, min.) | |
| Conductor to conductor | 2,000 |
| Conductor to shield | 1,000 |
| Component shield to all other shields | 500 |
| Overall shield to overall shield (type LS3SWUS only) | 200 |
| Insulation resistance (megohms/1,000 feet, min.) | |
| Conductor to conductor | 500 |
| Conductor to shield | 500 |
| Component shield to all other shields | 100 |
| Overall shield to overall shield (type LS3SWUS only) | 100 |
| Conductor continuity | No failure |
| Shield continuity | No failure |
| Jacket flaws (type LS3SWUS only) | No failure |



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| | |
|---|---|
| **Group A:** | |
| Visual and dimensional | No failure |
| Hydrostatic (open end) (leakage at 300 lb/in$^2$, 2 hours, in$^3$, max.) | See table I and table II |
| Capacitance | |
| Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet) | 30 |
| Capacitance unbalanced (percent, max.) | 10 |
| **Group B:** | |
| Cold bending (cable) | No failure |
| Thermoset proof test (percent, max.) | |
| Insulation | 50 |
| Jacket (when tested at 392 °F [200 °C]) | 50 |
| Drip (203±1.8 °F [95±1 °C]) | No failure |
| Physicals (unaged) | |
| Insulation (extruded insulation only) | |
| Tensile strength (lb/in$^2$, min.) | 700 |
| Elongation (percent, min.) | 150 |
| Jacket (cable) | |
| Tensile strength (lb/in$^2$, min.) | 1,300 |
| Elongation (percent, min.) | 160 |
| Tear (lb/in thickness, min.) | 35 |
| **Group C:** | |
| Physicals (aged) (air oven) | |
| Insulation (extruded insulation only) | |
| Tensile strength (percent of unaged, min.) | 75 |
| Elongation (percent of unaged, min.) | 75 |
| Jacket (cable) | |
| Tensile strength (percent of unaged, min.) | 60 |
| Elongation (percent of unaged, min.) | 60 |
| Permanence of printing (jacket) (cycles, min.) | 125 |
| Heat distortion (percent of unaged, max.) | 30 |
| Cable fill (sealant) removability | No failure |
| Shrinkage | No failure |



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|---|---|
| Shield (conformance to material, construction, and coverage) | No failure |
| Surface transfer impedance (type LS3SWUS only) | |
| Milliohms per meter (max.) | 700 |
| Electromagnetic pulse (EMP) response (dB, min.) | 60 |

**Group D:**

| | |
|---|---|
| Flame propagation (cable) | No failure |

**Qualification inspection:**

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

| | |
|---|---|
| Aging and compatibility (cable) (257±9 °F [125±5 °C]) | No failure |
| Abrasion resistance (jacket) (scrapes, min.) | 75 |
| Acid gas equivalent (percent, max.) | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Shield insulation | 2 |
| Halogen content (percent, max.) | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Insulation | 0.2 |
| Immersion (jacket) | |
| Tensile strength (percent of unaged, min.) | 50 |
| Elongation (percent of unaged, min.) | 50 |
| Smoke index (max.) | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 45 |
| Toxicity index (max.) | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |



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| | |
|---|---|
| Durometer (jacket) – type A (hardness, min.) | 80 |
| Weathering (jacket) | No failure |
| Electrical moisture absorption | No failure |

UNIT ORDERING LENGTHS:

| Types | Sizes | Feet (nominal) |
|---|---|---|
| LS3SWU and LS3SWUS | 3 through 19 | 1,000 |
| LS3SWU and LS3SWUS | 24 through 44 | 500 |

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

## CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-052)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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